



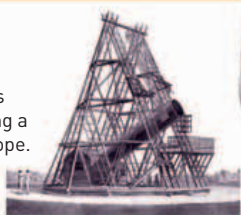
30MM THE DIAMETER OF THE **OBJECTIVE MIRROR** IN NEWTON'S TELESCOPE. TELESCOPES USED BY **MODERN ASTRONOMERS** HAVE MIRRORS UP TO **10,400MM** IN DIAMETER.

Newton's telescope

Isaac Newton made his first reflecting telescope in 1668. Shortly afterward, he made a second model (shown here), which stands about 8 in (20 cm) high. Newton's telescope was the first to use a primary mirror rather than a lens to collect light. A secondary mirror then reflects the light through a magnifying eyepiece for viewing.

1780s

William Herschel
Herschel discovers Uranus (1781) using a homemade telescope. He makes over 400 more, including a 1.26m reflector.



Herschel's 1.26m telescope



Hubble Space Telescope

1990–present

Space telescopes
Telescopes are put into space near Earth or orbit around it, from where they probe the sky in a range of wavelengths.

1920s

Edwin Hubble
Using the US's 2.5m Hooker telescope, Hubble shows that the Universe has more than 100 billion galaxies, and that it is expanding.



The Hooker telescope

1930s

Radio telescopes
A new field of astronomy—radio astronomy—begins when early radio telescopes detect radio waves from the Sun and distant galaxies.



Grote Reber's radio telescope

1960s–present

Exploring other worlds
Spacecraft are used to explore the Solar System. They fly past, orbit, and land on planets, moons, asteroids, and comets.



Mars rover